Week 6: Repeated Games

Cooperation, Punishment, and the Value of the Future

Chiu Yu Ko

CUHK Business School

Semester 2026–27 Term 1

Future interaction can change what is rational today.

From Week 5 to Week 6

Week 5: Commitment

- Act now
- Change future incentives
- Influence a later response
- Restrict future flexibility

Week 6: Repetition

- Play a stage game repeatedly
- Observe past behaviour
- Reward cooperation
- Punish deviation

Commitment changes the future game; repetition makes the future discipline the present.

Opening Decision: Price Competition Again

The Stage Game

Two firms simultaneously choose **Maintain** or **Discount** in every period.

	Rival: Maintain	**Rival: Discount**
You: Maintain	$(4, 4)$	$(1, 5)$
You: Discount	$(5, 1)$	$(2, 2)$

- In a one-shot game, Discount is strictly dominant.
- Discount–Discount is the unique Nash equilibrium.
- Maintain–Maintain gives both firms a higher payoff.

Can future interaction sustain Maintain–Maintain today?

First payoff: row player

Second payoff: column player

Today's Learning Outcomes

By the end of class, you should be able to:

1. distinguish a stage game from a repeated game;
2. explain why a known final period unravels cooperation;
3. compare finite, indefinite, and infinite horizons;
4. describe grim trigger and tit for tat;
5. interpret the discount factor δ;
6. calculate the minimum δ that sustains cooperation.

Stage Game Versus Repeated Game

Stage Game

The strategic interaction played in one period.

- Same players
- Same available actions
- One-period payoffs

Repeated Game

The stage game is played again over time.

- Past actions are observed
- Current actions affect future responses
- Total payoff includes future periods

The linkage across periods creates reputation, reward, and punishment.

Why a Known Final Period Unravels Cooperation

1. In the final period, there is no future punishment, so both firms Discount.
2. One period earlier, both know the final period will end in Discount–Discount.
3. Therefore, maintaining today cannot protect tomorrow's cooperation.
4. The same reasoning moves backward to the first period.

Backward-Induction Result

In a finitely repeated Prisoner's Dilemma with a known final period, the one-shot outcome repeats in every period.

Three Ways to Model the Horizon

Horizon	Meaning	Strategic Implication
Finite, known end	Everyone knows the final period	Last-period logic can unravel cooperation
Indefinite	Interaction continues next period with some probability	No certain final period; future discipline may matter
Infinite	A theoretical model with no terminal period	Convenient representation of an ongoing relationship

The key is not literal immortality; it is the absence of a known final interaction.

Two Repeated-Game Strategies

Grim Trigger

- Begin by cooperating.
- Continue while nobody deviates.
- After any deviation, punish forever.

Tit for Tat

- Begin by cooperating.
- Copy the rival's previous action.
- Punishment can end after cooperation resumes.

Trade-off

Grim trigger creates a stronger deterrent; tit for tat is more forgiving and may recover more easily from mistakes.

The Discount Factor δ

Interpretation

The discount factor measures how strongly a player values future payoffs relative to current payoffs.

- $0 \leq \delta \leq 1$.
- A larger δ means greater patience or a more likely continuation.
- A smaller δ means short-sightedness or an uncertain future relationship.
- A payoff next period worth 4 today contributes 4δ.
- A payoff two periods later contributes $4\delta^2$.

Cooperation is easier when the future matters more.

The Value of Cooperating Forever

Maintain–Maintain Gives 4 Every Period

$$V_C = 4 + 4\delta + 4\delta^2 + 4\delta^3 + \cdots$$

Solving for geometric sum:

$$\boxed{V_C = \frac{4}{1-\delta}}$$

- The first 4 is today's payoff.
- Future 4s are discounted by increasing powers of δ.
- This is an infinite geometric series.

The Value of Deviating Once

One-Time Gain Followed by Grim Punishment

A unilateral Discount gives 5 today. From the next period onward, both firms Discount and the deviator receives 2 each period.

$$\begin{aligned} V_D &= 5 + 2\delta + 2\delta^2 + 2\delta^3 + \cdots \\ &= 5 + \frac{2\delta}{1-\delta}. \end{aligned}$$

Temptation is immediate; punishment is delayed.

When Can Grim Trigger Sustain Cooperation?

No-Deviation Condition

Cooperation is sustainable when its value is at least as large as the value of deviating:

$$\frac{4}{1-\delta} \geq 5 + \frac{2\delta}{1-\delta}.$$

Solving for δ:

$$\boxed{\delta \geq \frac{1}{3}}$$

Interpretation

If the future is important enough, the loss of future cooperation outweighs the one-period gain from discounting.

Why Focus on Grim Trigger?

- It uses the harshest simple punishment in the pricing game.
- A harsher punishment can sustain cooperation at a lower discount factor.
- Therefore, grim trigger provides a clean benchmark for the minimum patience needed under severe punishment.
- A forgiving rule changes the recovery path; its threshold must be calculated rather than assumed.

But Harshness Has a Cost

If actions are observed imperfectly, permanent punishment can destroy cooperation after a mistake or false signal.

Monitoring and Recovery

Monitoring Problem

- Secret discounts may be hidden.
- Low demand may resemble cheating.
- Quality or effort may be difficult to verify.

Recovery Problem

- Permanent punishment may be excessive.
- Forgiveness helps recover from errors.
- Measurable performance supports credible responses.

Effective repeated relationships need observation, attribution, and a workable path back to cooperation.

Application 1: Long-Term Buyer–Supplier Relationship

Group Task

Model a buyer choosing Pay on Time or Delay Payment and a supplier choosing High Quality or Low Quality.

Build the stage game:

- Explain all four payoff cells.
- Identify the cooperative outcome.
- Identify each player's temptation.
- Identify the punishment outcome.

Analyse repetition:

- Calculate the minimum δ.
- Use two conditions if payoffs are asymmetric.
- Identify which player is more likely to deviate.
- Compare the threshold with the business setting.

Application 2: Asymmetric Relationships

Why Two Conditions May Be Needed

A powerful buyer and a dependent supplier may have different cooperative payoffs, temptation gains, punishment payoffs, and discount factors.

- Calculate the buyer's no-deviation condition.
- Calculate the supplier's no-deviation condition.
- Cooperation requires both conditions to hold.
- The player with the higher threshold is harder to discipline.

Managerial Question

Which side has more to gain from cheating now, and which side has more to lose from ending the relationship?

Checklist for a Strong Repeated-Game Application

Build the Model

- Real players and repeated action
- Defensible stage-game payoffs
- Clear cooperative and punishment outcomes
- Monitoring assumptions stated

Solve and Interpret

- Cooperation and deviation values correct
- Minimum δ calculated
- Asymmetry handled when relevant
- Threshold compared with the real relationship

Final Check

Can you explain where the payoff numbers come from and why the proposed punishment is credible and observable?

Application Debrief

1. What is the one-period temptation to deviate?
2. What future payoff is lost after deviation?
3. Which player requires the larger discount factor?
4. Is the proposed punishment credible and sufficiently observable?
5. Would forgiveness improve the relationship under noisy information?

Managerial Lesson

Cooperation depends on both incentives and institutions: future value, monitoring, punishment, and recovery.

Optional Extension: Design a Repeated Relationship

Use If Time Allows, or Continue Next Class

Select a real repeated relationship, such as retailer–supplier, platform–creator, employer–contractor, government–vendor, or competing firms.

1. Construct and defend the stage-game payoff matrix.
2. Specify grim trigger or a more forgiving response rule.
3. Calculate the minimum discount factor for every tempted player.
4. Explain monitoring errors and how cooperation can recover.
5. Recommend one change that lowers the required threshold.

Week 6: Main Takeaways

1. A repeated game links today's action to future responses.
2. A known final period can unravel cooperation through backward induction.
3. An indefinite or infinite horizon allows future discipline to matter.
4. Grim trigger is harsh; tit for tat is more forgiving.
5. The discount factor captures patience and continuation value.
6. Cooperation is sustainable when its discounted value exceeds deviation value.
7. Monitoring and recovery determine whether the strategy works in practice.

Define the stage game. Specify punishment. Compare future values. Interpret the threshold.

Exit Ticket and Looking Ahead

Exit Ticket

Why does a known final period undermine cooperation? In the pricing game, what does $\delta \geq 1/3$ mean economically?

Managerial Reflection

What can a firm change to make cooperation easier without relying on an excessively harsh punishment?

Supplementary: Deriving the Pricing Payoffs

Maintain outcomes

- Maintain–Maintain, $(4, 4)$: both protect margin and share demand.
- Maintain–Discount, $(1, 5)$: the discounter captures customers; the maintainer loses demand.

Discount outcomes

- Discount–Maintain, $(5, 1)$: the row firm captures customers through a unilateral discount.
- Discount–Discount, $(2, 2)$: both protect volume but sacrifice margin.

The values are illustrative rankings; a real application should derive them from price, demand, margin, and customer switching.

Supplementary: Geometric-Series Derivation

$$V_C = 4 + 4\delta + 4\delta^2 + \cdots,$$
$$\delta V_C = 4\delta + 4\delta^2 + 4\delta^3 + \cdots.$$

Subtracting gives

$$(1-\delta)V_C = 4, \qquad \boxed{V_C = \frac{4}{1-\delta}}.$$

General Formula

For $|r| < 1$,

$$a + ar + ar^2 + \cdots = \frac{a}{1-r}.$$

Supplementary: Solving the Threshold Step by Step

$$
\begin{aligned}
\frac{4}{1-\delta} &\geq 5 + \frac{2\delta}{1-\delta} \\
4 &\geq 5(1-\delta) + 2\delta \\
4 &\geq 5 - 3\delta \\
3\delta &\geq 1 \\
\delta &\geq \frac{1}{3}.
\end{aligned}
$$

Interpretation

The continuation value must be large enough to offset the immediate gain of $5 - 4 = 1$.

Supplementary: General Grim-Trigger Threshold

Let

R be the cooperative payoff, T the one-period temptation payoff, and P the punishment payoff.

$$\frac{R}{1-\delta} \geq T + \frac{\delta P}{1-\delta} \iff \delta \geq \frac{T-R}{T-P}.$$

- Cooperation is easier when R is larger.
- Cooperation is harder when the temptation gain $T - R$ is larger.
- A lower punishment payoff P strengthens deterrence.

Scope

This formula applies to grim trigger with a constant punishment payoff. A different response rule requires its own continuation-value calculation.

Supplementary: Symmetric and Asymmetric Thresholds

Symmetric Game

Both players have the same R, T, and P.

$$\delta \geq \frac{T-R}{T-P}.$$

One condition checks both players.

Asymmetric Game

Calculate separately:

$$\delta_1 \geq \frac{T_1-R_1}{T_1-P_1},$$

$$\delta_2 \geq \frac{T_2-R_2}{T_2-P_2}.$$

Both players' conditions must hold for mutual cooperation to persist.

Supplementary: Grim Trigger Versus Tit for Tat

	Grim Trigger	**Tit for Tat**
Initial action	Cooperate	Cooperate
Response	Punish forever after deviation	Copy the rival's previous action
Strength	Strong deterrence	Forgiving and responsive
Weakness	Fragile under mistakes	May create retaliation cycles
Threshold	Lower under severe punishment	Often requires greater patience

Supplementary: Exact Tit-for-Tat Calculation

After One Unilateral Deviation

Exact Tit for Tat makes each player copy the rival's previous action. The original deviator's payoff stream becomes

$$5,\ 1,\ 5,\ 1, \ldots$$

because play alternates between (D, M) and (M, D).

Continue cooperating

$$V_C = \frac{4}{1-\delta}$$

Deviate under exact Tit for Tat

$$V_D^{TFT} = 5 + \delta + 5\delta^2 + \delta^3 + \cdots = \frac{5+\delta}{1-\delta^2}$$

$$V_C \geq V_D^{TFT} \quad \Longleftrightarrow \quad \boxed{\delta \geq \frac{1}{3}}$$

Supplementary: What the Tit-for-Tat Check Establishes

Cooperative-Path Incentive Check

The calculation shows when a player prefers not to initiate a deviation from mutual cooperation.

Important Qualification

This condition alone does not establish that exact Tit for Tat is subgame perfect. Incentives must also be checked after asymmetric histories and along the alternating punishment path.

- Exact Tit for Tat may create repeated retaliation after one deviation or mistake.
- A reset or forgiveness convention changes the strategy and its payoff stream.
- Always specify the complete history-dependent response rule before calculating a threshold.

Supplementary: Common Errors

Common Error	Better Check
Repeat the one-shot matrix without a history-dependent strategy.	Specify how future play changes after cooperation or deviation.
Ignore the final period in a finite game.	Solve the final period and work backward.
Add future payoffs without discounting.	Weight period t by δ^t.
Use one condition in an asymmetric relationship.	Check every player with an incentive to deviate.
Assume punishment is perfectly triggered.	State monitoring, attribution, and recovery assumptions.